

HESSIAN-SVD NONLINEAR CLOSED-ORBIT CORRECTION IN A CESR MODEL BASED ON SCIBMAD

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Abstract

Placeholder. The completed abstract will state the closed-orbit correction problem, the Hessian-SVD method, the correction experiments, and the demonstrated validity boundary. No correction result is claimed in this working scaffold.

INTRODUCTION

Placeholder. This section will motivate finite-amplitude closed-orbit correction, connect the problem to the previously established nonlinear response-error analysis, and define the contribution of the Hessian-SVD approach. The literature context and final paper outline remain to be added.

PROBLEM DEFINITION AND NOTATION

Placeholder. This section will define the detector orbit, corrector vector, linear response matrix, Hessian tensor, objective function, constraints, and normalization used by the correction study. It will also specify the CESR model and the exact nonlinear reference calculation shared by all comparisons.

HESSIAN-SVD FORMULATION

Placeholder. This section will derive the projected Hessian representation and its singular-value decomposition, including truncation, regularization, and the relationship to conventional linear-SVD orbit correction. Algorithmic details will be added only after the formulation has been implemented and verified.

CORRECTION ALGORITHM AND EXPERIMENTS

Placeholder. This section will define the correction workflow, baselines, evaluation states, convergence criteria, and performance metrics. Tables and figures will be inserted after the correction experiments are complete; no numerical correction performance is asserted here.

HIGHER-ORDER VALIDITY BOUNDARY

The Hessian correction model is quadratic in the corrector increment. Its useful operating range is therefore channel dependent whenever the first neglected odd contribution becomes comparable with the retained even contribution. An existing vertical-only signed-parity experiment provides a concrete model-order boundary that will be reused when selecting correction step sizes and deciding when exact nonlinear re-evaluation is required.

Let $r(\Delta k)$ denote the nonlinear orbit residual after subtracting the nominal linear prediction $J\Delta k$. Paired positive and negative inputs separate its even and odd parts,

$$\begin{aligned} r_{\text{even}}(\Delta k) &= \frac{r(+\Delta k) + r(-\Delta k)}{2}, \\ r_{\text{odd}}(\Delta k) &= \frac{r(+\Delta k) - r(-\Delta k)}{2}. \end{aligned} \quad (1)$$

The experiment used 100 fixed vertical-corrector directions at 27 input radii, for 5401 states including the nominal state. All states converged without fallback. In the vertical detector channel, the even residual followed the expected quadratic trend and the odd nonlinear residual followed a cubic trend. Their fitted mean rms trends crossed at $\rho = 1.31$, corresponding to an active-corrector rms of approximately $6.5 \mu\text{rad}$. At $\rho = 6.4$, the mean odd/even ratio was 4.91. The simultaneous horizontal control ratio was only 3.05×10^{-4} , demonstrating that the breakdown of a quadratic approximation cannot be assigned from input amplitude alone; the input and detector channels must also be specified.

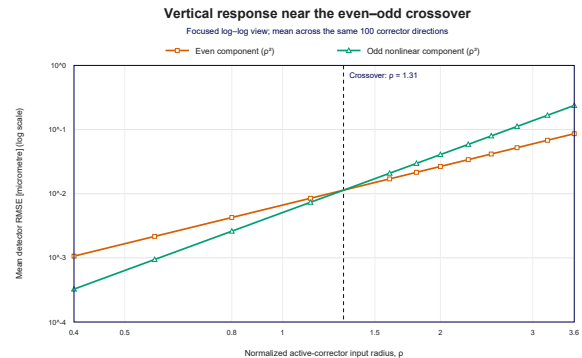


Figure 1: Mean vertical even and odd nonlinear residuals for vertical-only inputs. Their fitted quadratic and cubic trends cross at $\rho = 1.31$.

For the future correction experiments, this result is evidence for a channel-aware trust region rather than evidence that a particular Hessian-SVD algorithm succeeds. The completed study must test whether step restriction, iteration with exact nonlinear orbit re-evaluation, or a higher-order local model is required once the vertical odd component becomes significant. The physical source of the cubic contribution has not yet been identified.

CONCLUSION

Placeholder. The completed conclusion will summarize only verified correction performance and its model-order limits. At present, the transferred parity experiment establishes

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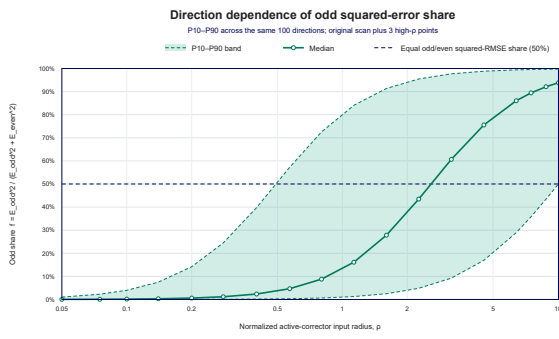


Figure 2: Odd squared-error share over the same 100 directions. The line is the median and the envelope spans P10–P90, showing strong direction dependence despite the small absolute error.

a candidate validity diagnostic, not a completed nonlinear orbit-correction result.